MISH 26
Excavation, Trenching, and Shoring

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MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

EXCAVATION, TRNCHING AND SHORING

Number

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1.0 PURPOSE

To define the responsibilities and requirements for protecting personnel engaged in excavation, trenching and shoring, and to ensure compliance with Occupational Safety and Health Administration/Washington Industrial Safety and Health Act (OSHA/WISHA) requirements.

2.0 RESPONSIBILITY

2.1 The Key Supervisor shall be responsible for the following:

2.1.1 Ensuring compliance with the provisions of this procedure, including training of employees.

2.1.2 Monitoring excavation, trenching and shoring operations.

2.1.3 Designating a “Competent Person”; ensuring this person has the training to act in this position; providing the competent person the authority to effectively discharge his/her duties.

2.2 The Competent Person(s) shall be responsible for the following:

2.2.1 Defining requirements and control measures.

2.2.2 Performing daily inspections to ensure compliance.

2.2.3 Stopping work when non-compliances or problems/concerns are identified from inspections.

2.3 Employees shall be responsible for compliance with this procedure.

3.0 DEFINITIONS

Actual Slope: The slope to which an excavation face is excavated.

Cave-In: The separation of a mass of soil or rock material from the side of an excavation, or loss of soil from under a trench shield or support system, and movement into the excavation in quantity that it could entrap, bury, injure, or immobilize a person.

Competent Person: One who can identify existing or predictable hazards in the surroundings that are unsanitary, hazardous, or dangerous to employees. Also have the authority by the nature of their positions to take prompt measures to eliminate the hazards. The person shall be knowledgeable in the requirements of this procedure.

Cross Braces: The horizontal members of a shoring system installed perpendicular to the sides of the excavation, the ends bear against either uprights or wales.

Distress: Soil in a condition where a cave-in is imminent or likely to occur. Distress indications may be fissures, slumping, spalling, raveling, or small amounts of materials separating from the face. The bottom may bulge or heave and the edge may sink or lower.

Equipment: Ladders, scaffolds, ramps, runaways, railings, barricades, sheet piling, shoring, bracing and any such safeguards. Protective construction and devices used in affording protection to the worker engaged in excavation work.

Embankment: An artificial or man-made bank of earthen material.

Excavation: Any man-made cavity or depression in the earth's surface, including its sides, walls, or faces, formed by earth removal and producing unsupported earth conditions by reasons of the excavation. If installed forms or similar structures reduce the depth-to-width relationship, an excavation may become a trench.

Faces or sides: The vertical or inclined earth surfaces formed because of excavation work.

Failure: The breakage, displacement, or permanent deformation of a structural member or connection to reduce its structural integrity and its supportive capabilities.

Maximum Allowable Slope: The steepest incline of an excavation face that is acceptable for the most favorable site conditions is the ratio of horizontal distance to vertical rise (H:V).

Moving Ground: Any ground, which for any reason, will not remain in its original location.

Protective System: A method of protecting employees from cave-ins, from material that could fall or roll from an excavation face or into an excavation, or from the collapse of adjacent structures. Protective systems include support systems, sloping systems, shield systems, and other systems that provide the necessary protection.

Registered Professional Engineer: A person that is registered as a Professional Engineer in the State of Washington in the appropriate discipline. The registered Professional Engineer shall comply with the Washington State Department of Licensing requirements, Chapter 18.43 RCW.

Sheeting: The members of a shoring system that retain the earth in position and in turn are supported by other members of the shoring system.

Shield (shield system): A structure that can withstand the forces imposed on it by a cave-in and thereby protect employees within the structure. Shields can be permanent structures or designed to be portable and moved along as work progresses. Shields can be remanufactured or job-built according to data from the manufacturer or designed by a registered Professional Engineer. Shields used in trenches are usually called "trench boxes" or "trench shields".

Shoring (shoring system): A structure such as a metal hydraulic, mechanical, or timber shoring system that supports the sides of an excavation that is designed to prevent cave-ins.

Sides, Walls, or Faces: The vertical or inclined earth surfaces formed because of excavation work.

Sloping (sloping system): A method of protecting employees from cave-ins by excavating to form sides of an excavation that are inclined away from the excavation to prevent cave-in.

Support System: A structure such as underpinning, bracing or shoring, which provides support to an adjacent structure, underground installation, or the sides of an excavation.

Trench: A narrow excavation made below the surface of the ground. The depth is generally greater than the width, but the width of a trench is not greater than15 ft.

Trench Jack: Screw or hydraulic type jacks used as cross bracing in a trench shoring system.

Unstable Rock: Rock material on the side or sides of the excavation not secured against caving-in or movement by rock bolts or by another protective system that has been designed by a registered professional engineer.

Unstable Soil: Earth material, which because of its nature cannot be depended upon to remain in place without extra support such as a system of shoring.

Wales: Horizontal members of a shoring system placed parallel to the excavation face whose sides bear against the vertical members of the shoring system or earth.

4.0 GENERAL REQUIREMENTS

4.1 Excavation, trenching, and shoring shall be conducted in accordance with the following statutory requirements:

• 29 Code of Federal Regulations (CFR) 1926, Subpart P; Excavations

• Washington Administrative Code (WAC) 296-155, Subpart N; Excavations, Trenching and Shoring

4.2 Employees in excavations shall be protected from cave-ins by an adequate protective system (sloping, shoring, or shields) in accordance with the provisions of subsection 5.4.

4.3 Soils should be tested/classified based on OSHA standard per subpart P appendix A

4.4 INSPECTIONS

4.4.1 Daily inspections of excavations, adjacent areas, and protective systems shall be conducted by a designated competent person to identify evidence of a situation that could result in possible cave-ins, for indications of failure of protective systems, and for hazardous atmospheres or other hazardous conditions. The competent person(s) shall be documented on the Competent Person Designation (see MISH 3).

4.4.2 Inspections shall also be conducted after every rainstorm or other potential hazard-increasing occurrence.

4.4.3 Inspections shall be performed using the "Daily Excavation/Trenches Safety Inspection Checklist" (attachment MISH 26-A).

NOTE: These inspections are only required when employee exposure can be reasonably anticipated (i.e.; 4 ft. or more in depth and occupied by employees).

4.4.4 Inspections shall be documented on the Daily Excavation/Trenches Safety Inspection Log (see attachment MISH 26-B)

4.4.5 When inspection reveals evidence of a situation that could result in a possible cave-in, indications of protective system failure, hazardous atmospheres, or other hazardous conditions, employees shall be removed from the hazardous area until the necessary precautions have been taken to ensure their safety.

4.5 Employees shall not work in excavations where water is accumulating or has accumulated unless adequate precautions have been taken to protect employees against the hazards posed by water accumulation.

4.6 Employees exposed to vehicular traffic shall be provided with and shall wear warning vests or other suitable garments marked with or made of reflectorized or high-visibility material.

4.7 When the configuration of an excavation is such that the excavation is deemed to be a confined space, the provisions of MISH 16 shall apply.

4.8 Tests should be conducted for air contaminants such as oxygen and flammable gases. Atmospheric testing should be completed before entry and periodically for the duration of operations.

4.9 Adequate barriers for physical protection shall be provided at all excavations. All wells, pits, shafts, etc. shall be barricaded or covered.

4.10 When mobile equipment is used or allowed next to excavations or trenches, stop logs or barricades shall be installed. Such devices shall not be required for equipment conducting the actual excavating or backfilling operations.

4.11 SPECIAL PRECAUTIONS FOR EXCAVATING/TRENCHING AROUND ELECTRICAL UTILITIES

4.11.1 Each underground electrical interface with new construction shall be independently reviewed on a case-by-case basis.

4.11.2 If it can be determined from installation as-builts that the energized cable is protected by a rigid steel conduit raceway or concrete encasement, then de-energizing the cable is not required and excavation with hand tools such as shovels, pry bars, picks or air lances is acceptable.

4.11.3 If it is determined from installation as-builts that the energized cable is protected by a PVC conduit raceway, then de-energizing the cable is not required and hand excavation as defined in paragraph 4.10.6 and subsection 5.2 is acceptable.

NOTE: De-energizing underground lines when excavating near or around them is always the safest and preferred method to mitigate electrical hazards during excavation.

4.11.4 If actual installation details are indeterminate, or the service is a direct buried cable not protected by rigid steel raceway, concrete encasement, or PVC, then the cable shall e de-energized.

EXCEPTION: If it is determined that the system cannot be de-energized due to the adverse effects on necessary operation or life safety systems, then justification to hand excavate must be provided. Such justification requires concurrence of the division manager, Industrial Safety and Health (IS&H).

4.11.5 Minimum safety precautions during hand excavations of an energized direct buried cable shall include:

• The excavation worker shall wear protective insulated gloves, rated for the voltage potential and meeting the specifications of ANSI/ASTM D120-1984.

• A qualified standby person shall be in attendance during excavation. The standby person shall be CPR certified, first-aid trained, and trained in emergency electrical practices.

• The suspected cable location shall be periodically verified using a hand-held inductance meter.

4.11.6 Hand excavation around unprotected, energized electrical utilities is defined as the removal of surface and subsurface materials, including sand, gravel, soil, and rocks by direct labor methods where the only acceptable hand assist tool is a common hand shovel. The use of requiring hand-held picks and pry bars is expressly prohibited in situations requiring hand excavation. The use of a hand shovel to stab the soil in an effort to loosen or penetrate excavation material with anything greater than normal foot pressure is prohibited. The only acceptable aid to a hand excavation operation is an air-assisted dielectric soil pick, (e.g., air lance). The air lance is permitted to loosen compacted soil and/or expose a buried utility where soil conditions are sufficiently compacted, rocky or the presence of an underground utility is suspected and not yet physically verified.

5.0 PROCEDURE

5.1 PREPARATION

5.1.1 Prior to excavation, permits (internal and external, when required by contract) shall be obtained from authorized organizations and included with pre-job planning documents where applicable. In addition, the necessary drawings, specification, and restrictions applicable to the excavation shall also be obtained.

5.1.2 Reference drawings and documents shall be checked in detail to determine location of all known underground obstructions, pipes, power lines, phone lines, foundations, etc. These locations shall be marked at the worksite in a conspicuous manner.

5.1.3 The Contractor shall contact the telephone utilities for underground phone lines.

5.1.4 Utilities and facilities shall be de-activated, de-energized and/or interrupted, as required, and the source locked and tagged in accordance to MISH 15. Coordination for this activity will be done by MH Construction Management.

5.1.5 Just prior to excavation, the responsible Key Supervisor shall brief the excavation crew as to any special requirements or obstructions in the excavation area. The excavation permit and a Survey Scanning Report (when required) shall be at the excavation site during excavation as a reference for the excavation crew.

5.1.6 A flagman is used, when required, to assist the equipment operator during excavation.

5.1.7 All surface items to be protected shall be made visible to the equipment operator by high-viz tape or by a 7-foot bicycle flag mounted above the object

5.1.8 The equipment operator shall be made aware of these protected items prior to operating equipment such as cranes, trucks, backhoes, or loaders.

5.1.9 A stairway, ladder, ramp or other safe means of egress shall be located in trenches hat are 4 ft. or more in depth so as to require no more than 25 ft. of lateral travel for employees.

5.2 HAND EXCAVATION

5.2.1 All known underground obstructions shall be exposed by hand excavation beginning at a distance of 3 ft. from the obstruction. For hand excavation around electrical utilizes, also see subsection 4.10.

NOTE: Machines may be used to remove the soil from excavation, but shall not be used to dig undisturbed soil unless utility is can be seen by operator.

5.2.2 In work areas where the exact location of underground electrical power lines in unknown, employees using jack hammers, bars, or other hand tools which may contact a line shall be provided with insulated gloves.

5.2.3 If obstructions shown on the drawings and/or permits are not found in the location indicated, or if obstructions are encountered that are not shown, the work in that area shall be stopped and MH Construction Management shall be contacted for direction as to how the work is to proceed.

5.3 MACHINE EXCAVATION: machine excavation, when permitted, shall take place after compliance with the above procedures. The following requirements apply:

5.3.1 MH Construction management shall be notified of damage to utilities or facilities and of any abnormal conditions encountered.

5.4 PROTECTIVE SYSTEMS

5.4.1 Excavations and protective systems shall be designed so that cave-ins and slides are avoided and safe conditions shall be maintained for employees working in and around the excavation.

5.4.2 Whenever practical and feasible, the sides of excavations and trenches shall be sloped at an angle not steeper than 1-1/2 horizontal to 1 vertical (see attachment MISH 26-C).

5.4.2.1 When sloping is not practical or feasible, the sides of excavations and trenches may be vertical if they are provided with support systems (shoring or shielding).

5.4.2.2 A support system shall be required in trenches more than 4 ft. deep under the following circumstances:

• The slopes are steeper than 1-1/2 horizontal to 1 vertical

• The slopes are at least 1-1/2 horizontal to 1 vertical but do not extend to the bottom of the trench

5.4.2.3 Support systems shall be installed and removed in a manner that protects employees from cave-ins and structural collapses and from being struck by members of the support system.

5.4.2.4 Excavation of material to a level no greater than 2 ft. below the bottom of the members of a support system shall be permitted only if the system is designed to resist the forces calculated for the full depth of the trench and there are no indications, while the trench is open, of a possible loss of soil from behind or below the bottom of the support system.

5.4.2.5 Shoring shall support the vertical portion of a trench and shall extend above the bottom of the slope at least 18 inches to prevent material from sliding into the trench. The surface of the slope shall be cleared of boulders, stumps, or other hard masses of earth.

5.4.2.6 A support system may not be required under the following circumstances:

• The trench is less than 4 ft/5ft. deep.

• The trench is less than 20 ft. deep and the slopes are at least 1-1/2 horizontal to 1 vertical and extend to the bottom of the trench.

5.4.3 Timber cross-brace or trench jacks shall be placed in a true horizontal position, vertically spaced, and secured to prevent sliding, falling, or kick outs. Wales shall be placed with the greater dimension horizontal.

5.4.4 The sides of an excavation next to a previously backfilled area shall be sloped at least one and one-half horizontal to one vertical, particularly when the separation is less than the depth of the excavation.

5.4.5 When employees or equipment must cross over an excavation 6ft in depth, a walkway or bridge with standard guardrails shall be provided.

5.4.6 Engineering-approved portable trench boxes r sliding trench shields may be used instead of a shoring system or slope. They shall protect as well as, or better than, the shoring required for the trench and are used in accordance with the manufacturer's recommendations.

5.4.7 Materials used for support systems shall be in good, serviceable condition. Sheeting and four-by-four inch timber cross-braces shall be Douglas Fir, No. 2, or better. Wales and six-by-six inch timber cross-braces shall be Douglas Fir, No. 1, or better

5.4.8 When employees are required to enter and excavation, excavated material shall be kept back at least two feet from the edge of the excavation.

5.4.9 When the slopes of a trench are steeper than one and one-half horizontal to one vertical, and engineering-approved trench boxes or shields are not used, shoring shall be placed as detailed in the OSHA or WISHA standards.

5.4.9.1 A combination using a simple slope with a vertical sided lower portion may be used ash shown in attachment MISH 26-A, Detail 2.

5.4.9.2 Mudsills (wales installed at the toe of the trench side, having the same size and cross bracing spacing as the wales directly above them) may be used in place of embedding the uprights.

5.4.10 Sloping or shoring designs for trenches and excavations deeper than 20 ft. shall have Registered Professional Engineering approval prior to placement.

5.4.11 Employees should not work under loads of digging equipment where loads may fall.

5.4.12 Shoring, bracing, or underpinning shall be provided to ensure the stability of adjoining structures, which may be endangered by an excavation. Among others, these structures would include the following:

• Buildings and walls

• Concrete encased raceways, piping, etc.

• Storage tanks

• Power poles

6.0 RECORDS

Document Record Submittal Responsibility Record Retention Responsibility Competent Person Designation

Contractor Construction Management Shoring Designs

Contractor Construction Management Daily Excavation/Trenches Safety Inspection Log

NA Contractor

7.0 REFERENCES

Not Applicable

8.0 ATTACHMENTS

MISH 26-A Daily Excavations/Trenches Safety Inspection Checklist MISH 26-B Daily Excavation/Trenches Safety Inspection Log MISH 26-C Excavation Detail Sketch

MH CONSTRUCTION EXCAVATION/TRENCHES INSPECTION CHECKLIST

A SUPERVISOR/COMPETENT PERSON SHALL MAKE DAILY SAFETY INSPECTIONS OF EXCAVATIONS/TRENCHES, ADJACENT AREAS, AND PROTECTIVE SYSTEMS FOR EVIDENCE OF SITUATIONS THAT COULD RESULT IN POSSIBLE CAVE-INS, INDICATIONS OF FAILURE OF PROTECTIVE SYSTEMS, HAZARDOUS ATMOSPHERES, AND OTHER HAZARDOUS CONDITIONS. THE SUPERVISOR/COMPETENT PERSON SHALL CONDUCT INSPECTIONS DAILY AND AFTER OTHER POTENTIAL HAZARD INCREASING OCCURRENCES (HEAVY RAIN, ETC.). THESE INSPECTIONS ARE REQUIRED ONLY WHEN WORKER EXPOSURE CAN BE REASONABLY ANTICIPATED IN EXCAVATIONS OVER 4 FEET DEEP AND TO BE OCCUPIED BY WORKERS.

VERIFY THE FOLLOWING ITEMS (AS APPLICABLE) & DOCUMENT RESULTS ON THE INSPECTION LOG.

• WORKERS HAVE BEEN BRIEFED ON EXCAVATION HAZARDS RELATED TO WORK.

• AREA SURROUNDING EXCAVATION HAS BEEN PROTECTED AND BARRICADED.

• UTILITIES HAVE BEEN LOCATED, PROTECTED AND MARKED.

• EXCAVATIONS OVER 4 FEET DEEP ARE PROPERLY SLOPED/SHIELDED/SHORED.

• FOUNDATIONS, STRUCTURES, APPURTENANCES, ETC., ARE SUPPORTED.

• PROTECTIVE SYSTEMS USED ARE IN ACCORDANCE WITH MFR. SPECS/ DESIGN.

• SLOPING/PROTECTIVE SYSTEMS HAVE BEEN CHECKED FOR EVIDENCE OF FAILURE (SOIL DISTRESS, SOIL FISSURES, STRUCTURAL DAMAGE, ETC.)

• A MEANS OF WORKER ESCAPE (STAIRWAY, LADDER, RAMP) IS AVAILABLE WITHIN 25 FEET LATERAL TRAVEL OF ALL WORKERS IN THE EXCAVATION.

• WALKWAYS OR BRIDGES, WITH STANDARD GUARDRAILS, ARE PROVIDED WHERE PERSONS ARE REQUIRED TO CROSS THE EXCAVATION/TRENCH.

• EXCAVATION SPOILS & MATERIALS HAVE BEEN PLACED AT LEAST 2 FEET AWAY FROM THE EDGE OF THE EXCAVATION/TRENCH.

• DETERMINATION HAS BEEN MADE AS TO WHETHER THE EXCAVATION/TRENCH IS CONSIDERED A CONFINED SPACE (NON-PERMIT OR PERMIT REQUIRED).

• THE EXCAVATION/TRENCH IS SAFE FOR WORKERS TO ENTER.

• CHANGES IN WORK SCOPE, SITE CONDITIONS OR HAZARDS HAVE BEEN REVIEWED WITH EXCAVATION WORKERS, AS THEY HAVE OCCURRED.

• CORRECTIVE MEASURES HAVE BEEN TAKEN TO ELIMINATE EXISTING AND PREDICTABLE HAZARDS.

MH CONSTRUCTION DAILY EXCAVATION/TRENCHES SAFETY INSPECTION LOG/CHECKLIST

APPLICABLE ITEMS LISTED ON THE INSPECTION CHECKLIST WERE VERIFIED AND DEFICIENCIES ARE NOTED WITH THE CORRESPONDING CORRECTIVE ACTION THAT WAS TAKEN TO ELIMINATE THE HAZARD.

DATE TIME COMMENTS/DEFICIENCIES NOTED CORRECTIVE ACTION TAKEN INSPECTOR

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